CICSA PROJECT

WAZUH SETUP & LOG COLLECTION

Red Team Hacker Academy

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ABSTRACT

this project focuses on deploying and configuring Wazuh, an open-source SIEM platform, to collect, monitor, and analyze security logs from Kali Linux systems in real time.

The setup involves installing Wazuh Manager as the central monitoring server and configuring the Wazuh Agent on Kali Linux to securely forward system, authentication, and security event logs. Wazuh's features, such as log analysis, intrusion detection, and file integrity monitoring, enable efficient threat detection and compliance management.

By implementing this configuration, the project establishes a secure and centralized log collection framework, enhancing visibility into the system's security posture. It demonstrates how open-source SIEM tools like Wazuh can support proactive monitoring, incident response, and security analysis in Linux environments.

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INTRODUCTION

In today's cybersecurity landscape, monitoring and analyzing system logs are essential for detecting and responding to security incidents. Logs provide critical insights into system activities, authentication events, and potential threats, making centralized log management vital for security and compliance.

Wazuh, an open-source SIEM tool, enables real-time monitoring, log analysis, and intrusion detection, integrating with the Elastic Stack for visualization and alerting through the Wazuh Dashboard.

This project, "Wazuh Setup & Log Collection in Kali," focuses on deploying and configuring the Wazuh Manager and Agent to collect and analyze logs from Kali Linux. The Manager acts as the central server for processing logs, while the Agent forwards system logs securely for centralized monitoring.

By implementing this setup, the project establishes a secure, efficient log collection framework that enhances visibility, detects unusual activity, and strengthens the overall security posture of the Kali Linux environment.

PREREQUISITES

HARDWARE REQUIREMENTS

 Processor: Dual-Core CPU or higher (The Wazuh central components require a 64-bit Intel, AMD, or ARM Linux processor (x86_64/AMD64 or AARCH64/ARM64 architecture) to run)

 RAM: Minimum of 4 GB (Recommended-8 GB)

 Storage: At least 50 GB free disk space

SOFTWARE/NETWORK REQUIREMENTS

Wazuh recommends any of the following operating system versions:

- Ubuntu 16.04, 18.04, 20.04, 22.04, 24.04
- Amazon Linux 2, Amazon Linux 2023
- CentOS Stream 10
- Red Hat Enterprise Linux 7, 8, 9, 10
- Network: Stable connection between Wazuh Manager and Agent

Wazuh Components:

Wazuh Manager	Main system receiving and analysing logs
Wazuh Agent	Installed on endpoints to collect and send system logs and security events to the manager.
Wazuh Dashboard	Web interface for monitoring and configuration
Wazuh Indexer	Stores and indexes event data for fast searching and analysis.

PROCEDURE

INSTALLATION OF WAZUH MANAGER

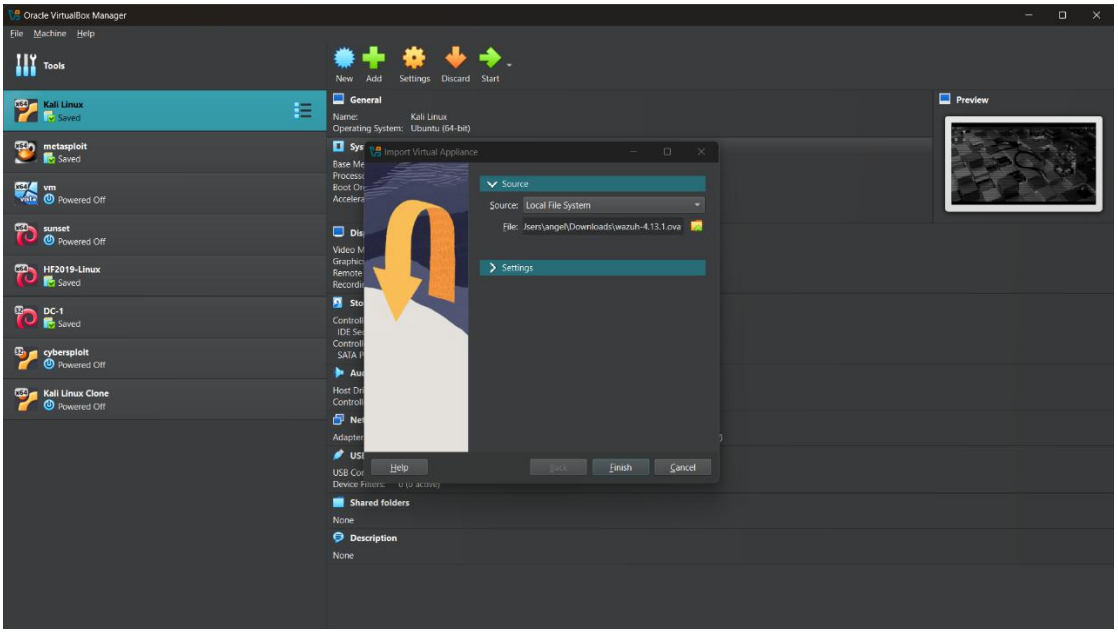
Step 1: Download Wazuh Manager package/ova file using:

https://packages.wazuh.com/4.x/vm/wazuh-4.13.1.ova

step 2: Import OVA into VirtualBox

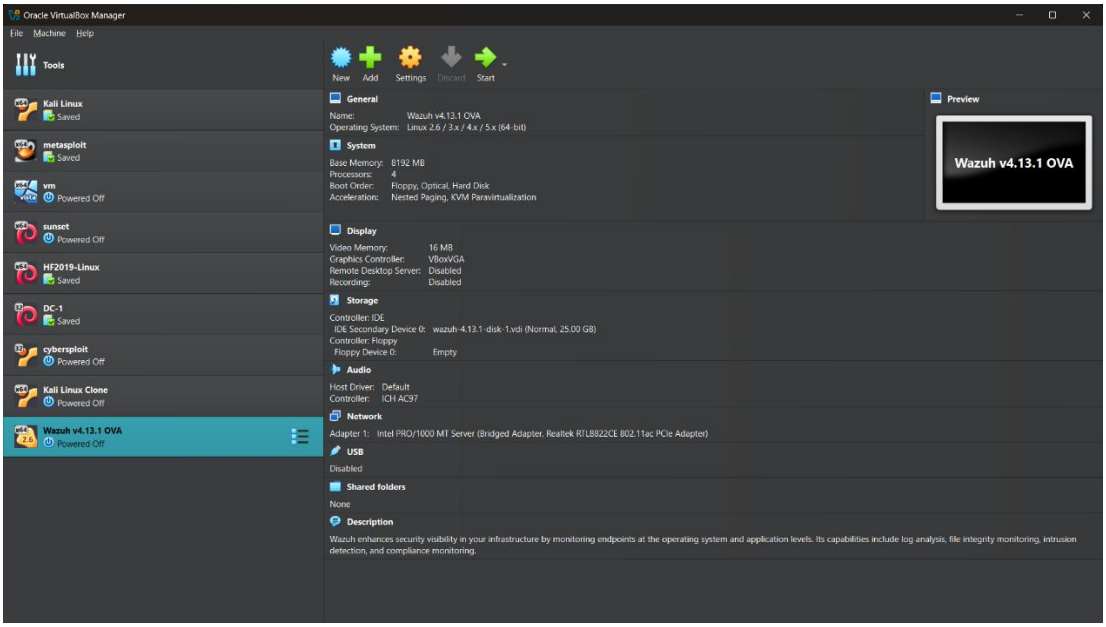
Open VM and from the top menu: Go to File → Import Appliance.

In the Import dialog: click the folder icon and select the downloaded wazuh-4.13.1.ova file, Review the VM settings and click Import.



Step 3: After import, configure these VM settings to match your host capacity and start the machine.

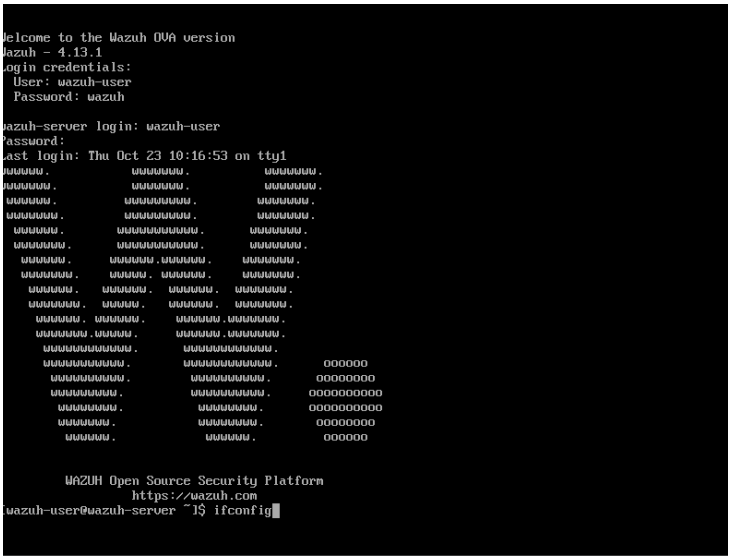
Base Memory (RAM) - Recommended: 4096–8192 MB (4–8 GB), Processor - 2–4 CPUs, Network – Bridge Adapter

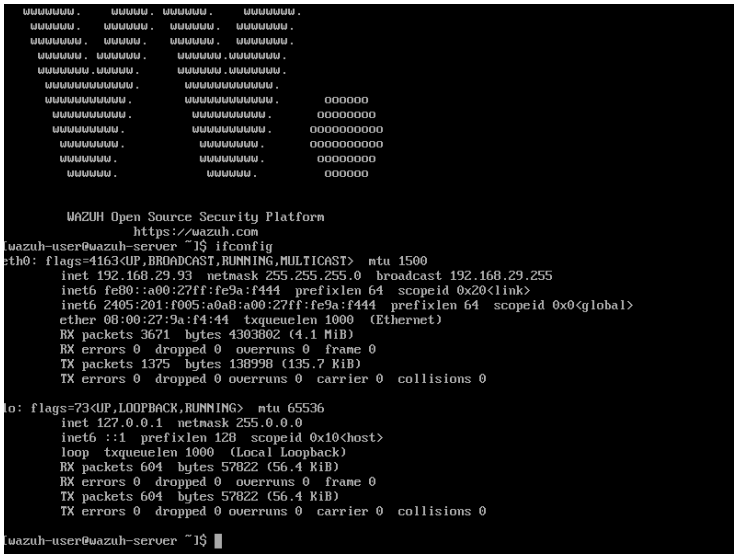


Step 4: Start the imported Wazuh Manager VM

Log in using the default username/password provided

Run the command ifconfig and note down the IP



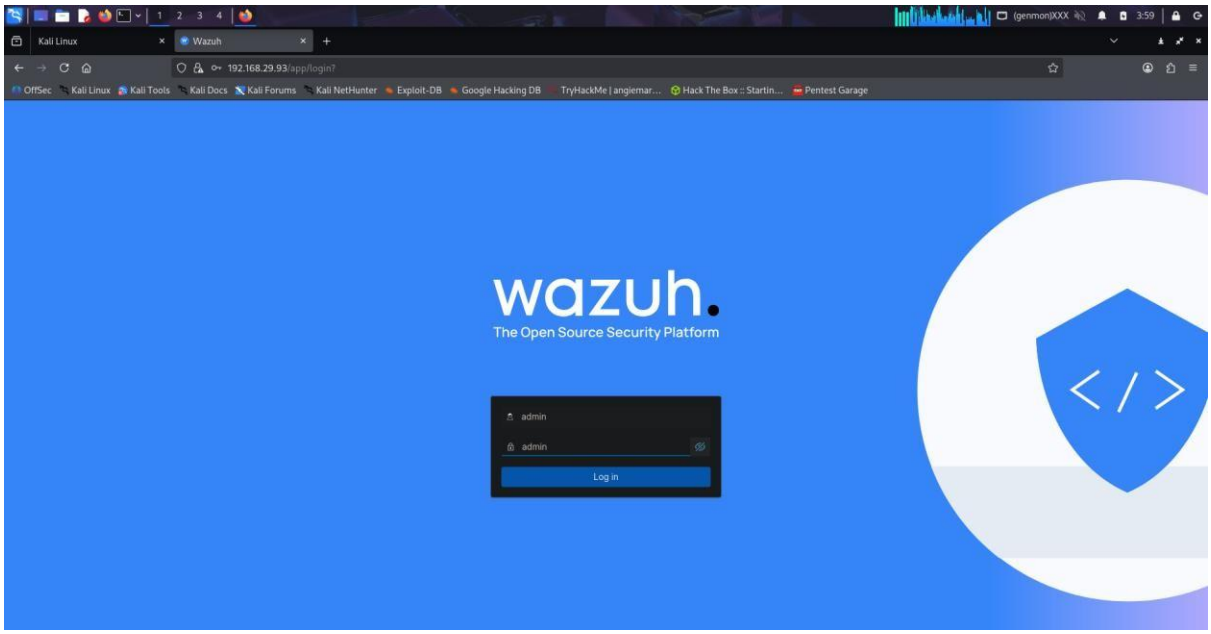


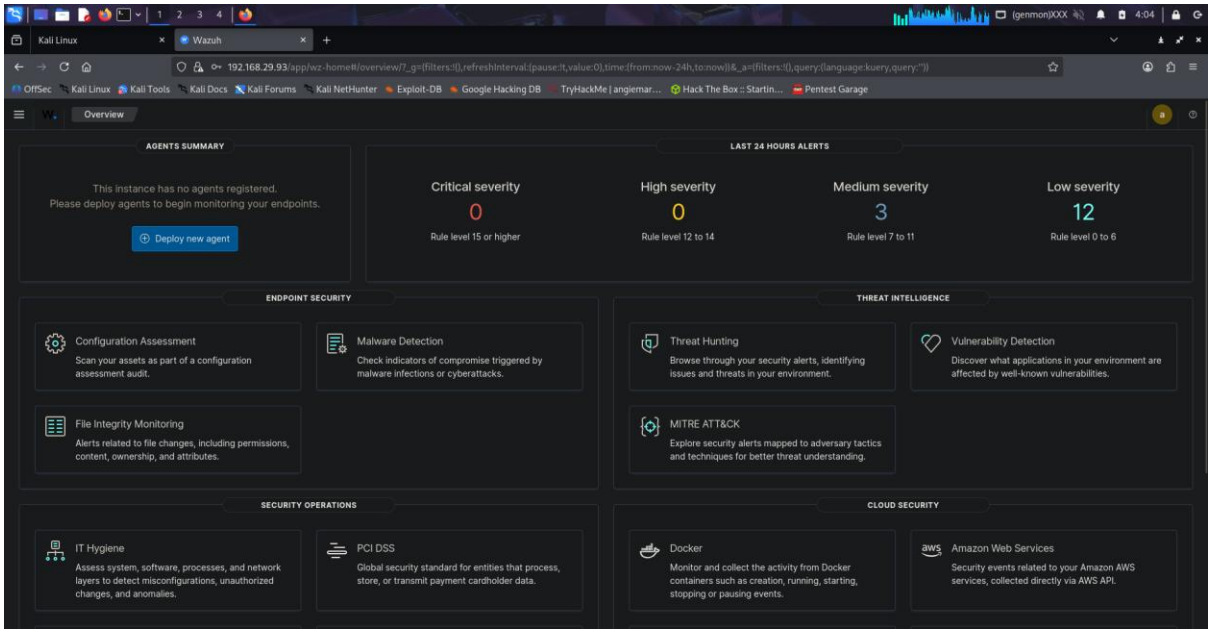
Step 5: Open the web browser on the Kali Linux VM and navigate to the Wazuh Webpage using the IP address.

Login to Wazuh with:

Username: admin

Password: admin

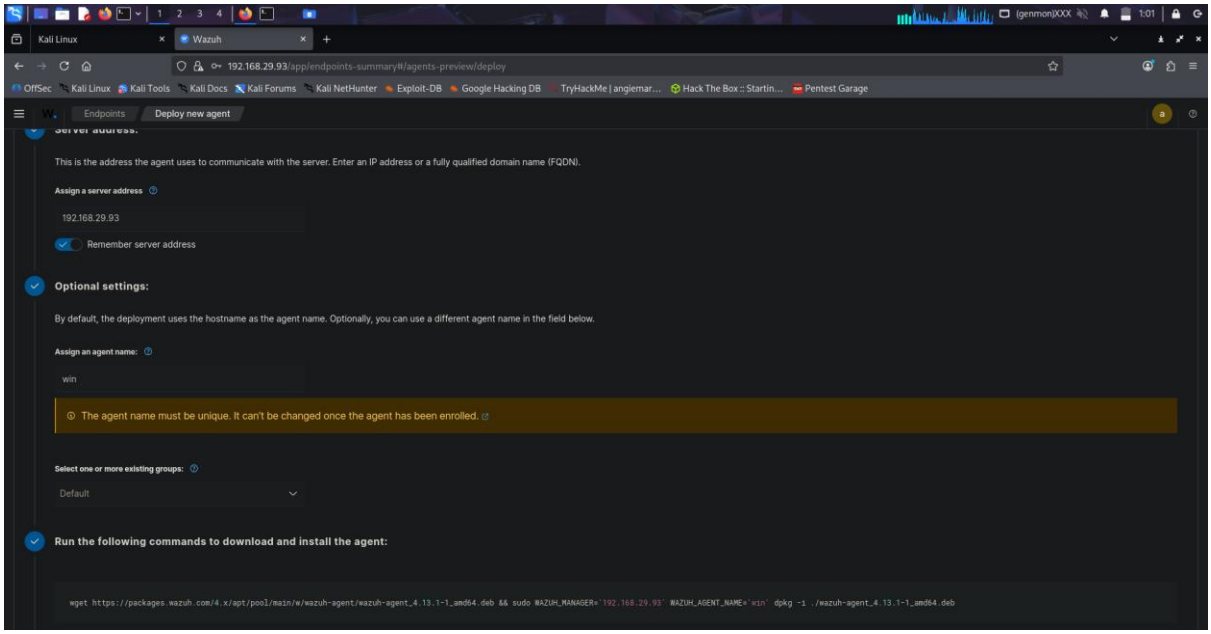


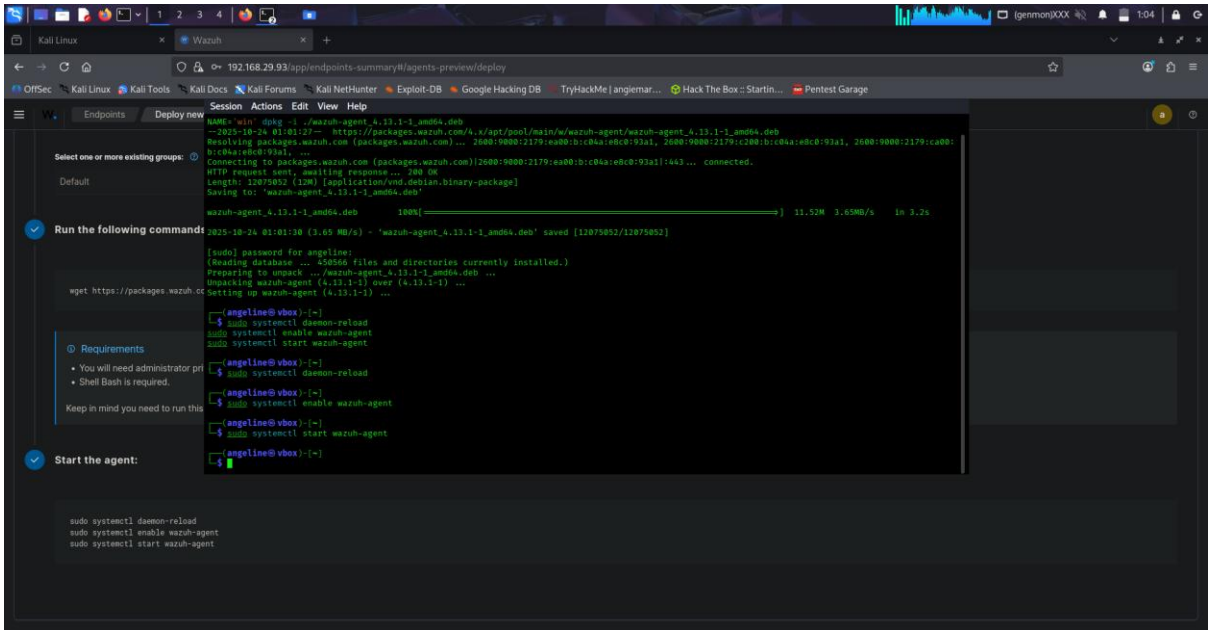


WAZUH AGENT SETUP ON KALI LINUX

Step 1: Navigate to the Wazuh Dashboard and click deploy new agent and fill in the details.

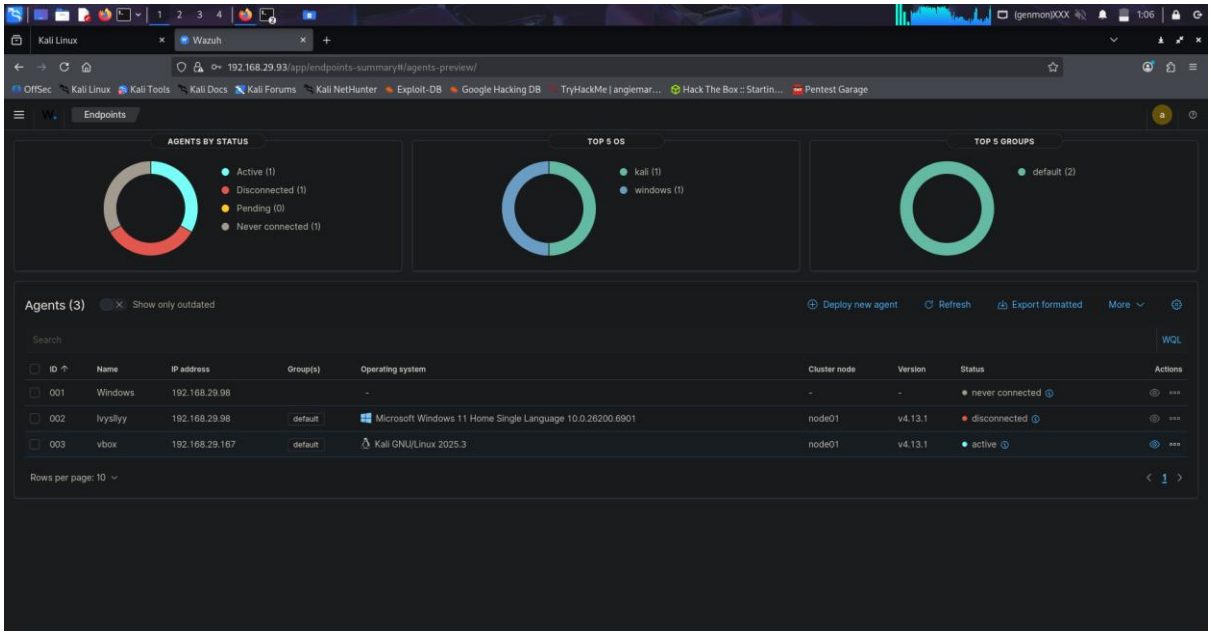
Copy-paste the given commands into the terminal and run it.





Step 2: Check the agent status

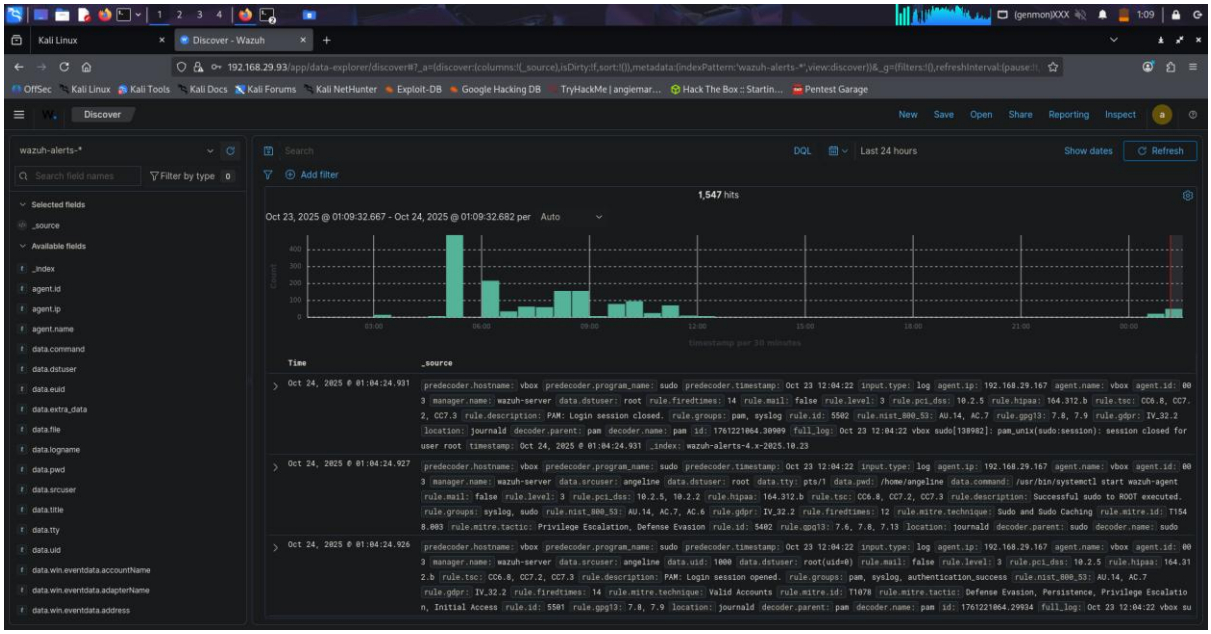
Check whether the Deployed agent status is active.

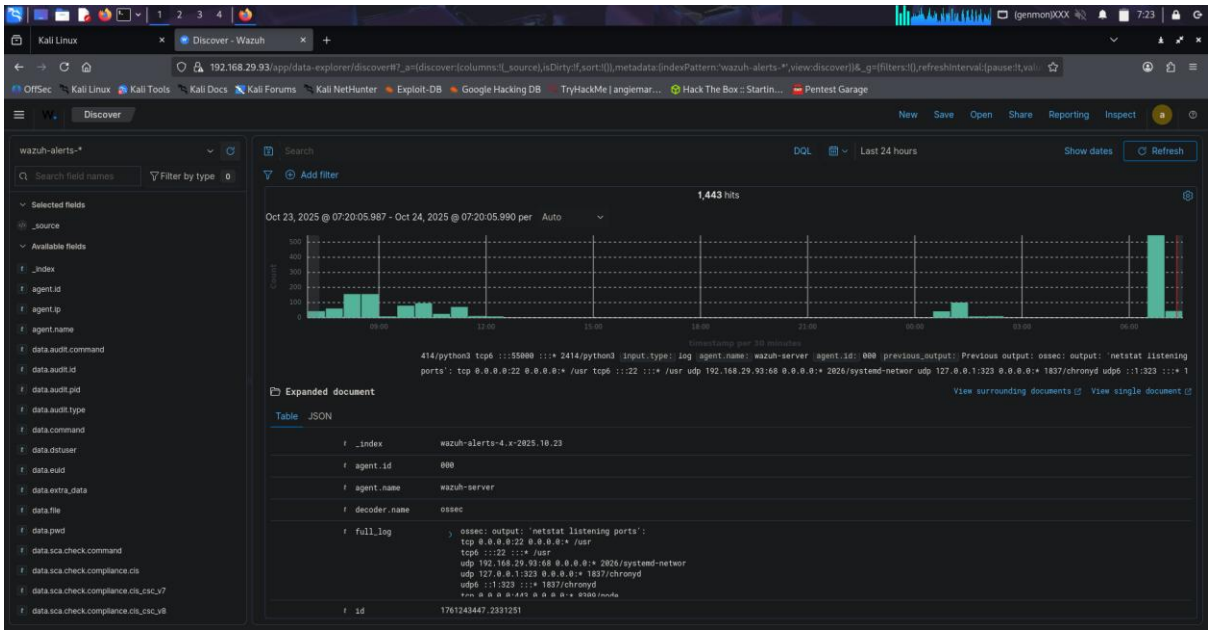


The agent's Active status in the Wazuh dashboard shows that it has been successfully deployed as expected and is effectively communicating with the Wazuh Manager.

LOG COLLECTION AND VERIFICATION

The collected logs were verified by navigating to Explore → Discover in the Wazuh dashboard, where the logs were successfully displayed for review.





The logs captured system activities such as user logins, command executions, and security events from the Kali Linux machine, were successfully transmitted to the Wazuh Manager for analysis and categorization, confirming that the agent was correctly configured and communicating effectively with the manager.

CHALLENGES FACED

During this project, I encountered several technical challenges while setting up and connecting the Wazuh Manager and Agent. Installing and configuring both systems required careful attention, as minor mistakes with IP addresses, ports, or configuration settings could prevent them from communicating properly.

Generating and verifying logs from Kali Linux sometimes required extra steps, and viewing real-time events on the Wazuh dashboard was occasionally delayed. Troubleshooting these issues often involved checking configuration files and restarting services multiple times, but through trial, error, and careful review, I was able to resolve the problems and successfully complete the setup.

CONCLUSION

The project focused on efficiently collecting and monitoring Linux system logs using the Wazuh platform. While implementing it, I learned how to install and configure both the Wazuh Manager and Agent to enable smooth log transfer and centralized monitoring. By ensuring proper communication between the systems, I was able to verify that the logs were successfully collected and displayed in the Wazuh dashboard. I also explored how Wazuh analyzes and visualizes security events, providing valuable insights.

Overall, this project helped me understand the importance of log management and system monitoring, while also improving my practical skills in using Wazuh as a SIEM tool.

REFERENCES

https://youtu.be/bltbJ2TUQWU?si=fBZLM9ADq1ue3mTw

https://youtu.be/u2pS0Zuzbmc?si=fYw_kSQvjVWaTS3X